



UCAC

Université Catholique
d'Afrique Centrale

FACULTE DE SCIENCES SOCIALES ET DE GESTION

CAMPUS DE NKOLBISSON

Examen Final : Algorithmics and Programming

Duration: 2 hours; Class: BEM 2; Teacher: Dr Loumngam

Instructions for the Student:

Objective: This exam will test your understanding of the fundamental concepts of Python programming as covered in the course.

Submission Guidelines:

1. You will be provided with a Python file named `exam_submission.py`.
2. Answer all theoretical questions by writing your answers in comments (#) in the designated areas (YOUR_ANSWER_HERE).
3. Answer all practical questions by completing the Python functions provided in the file.
4. Do not change the names of the functions or the structure of the file.
5. Once you have completed the exam, save the `exam_submission.py` file and submit it.

Exam Paper: Introduction to Python Programming

Total Points: 100

Section 1: Theoretical Knowledge (30 Points)

Question 1: Valid Identifiers (5 Points)

From the list below, identify which are valid Python identifiers and which are invalid. For each invalid identifier, briefly explain why, based on the rules for naming objects.

- `First_Initial`
- `1stprogram`
- `down.to.earth`
- `see you`
- `for`
- `_my_variable`

Question 2: Data Types and Mutability (10 Points)

For each of the following data definitions, identify the specific data type (e.g., int, str, list, bool) and state whether that data type is mutable or immutable.

- `a = "Hello Python"`
- `b = 14_000_000`
- `c = ['honda', 'yamaha', 'suzuki']`
- `d = False`
- `e = 19.25`

Question 3: Core Concepts Explanation (9 Points)

In your own words, but based on the what we did in class, briefly explain the following concepts:

- **Dynamic Typing:** What does it mean that Python is a dynamically typed language?
- **List Comprehension:** What is the primary advantage of using a list comprehension compared to a for loop?
- **The with statement for files:** Why is it recommended to use the with statement when opening files?

Question 4: IDLE Environment (6 Points)

Distinguish between the IDLE Shell and the IDLE File Editor. What is the primary use case for each?

Section 2: Code Analysis (20 Points)

Question 5: Variable Assignment (8 Points)

What will be the final values of variables A and B after the execution of the following instructions? Explain the state of each variable after each line of code.

`A = 5`

`B = 4`

`C = A - B`

`A = 2 * C`

`B = C - A`

Question 6: List Manipulation (12 Points)

What is the final state of the `guests` list after all operations are executed? Explain what each method (`.append()`, `.insert()`, `.pop()`, `.remove()`) does to the list at each step.

`guests = ['Newton', 'Einstein', 'Copernicus']`

`guests.append('Galilei')`

`guests.insert(0, 'Kepler')`

```
guests.pop()
```

```
guests.remove('Einstein')
```

Section 3: Practical Application (50 Points)

Question 7: String Manipulation (15 Points)

Write a Python function `format_user_message` that takes a person's name as a variable. The function should then return a single string message formatted as: *"Hello [Name], would you like to learn some Python today?"*, where [Name] is the person's name in title case.

Question 8: Lists and Loops (20 Points)

Write a Python function `process_inventory` that takes a list of products (strings). The function should perform the following actions:

1. Use a for loop to print a statement for each product, such as *"I have [product] in stock."*.
2. Permanently sort the list in alphabetical order.
3. Remove the item 'bread' from the list.
4. Return a formatted string indicating the number of items remaining in the inventory, for example: *"There are 5 items left in inventory."*

Question 9: File I/O (15 Points)

Write a Python function `create_and_read_report` that performs the following steps:

1. Creates and opens a file named `report.txt` in write mode ('w') using a context manager (with statement).
2. Writes the following two lines to the file:
 - o *"Line 1: System status is nominal.\n"*
 - o *"Line 2: All modules are online.\n"*
3. The file should be closed automatically after writing.
4. The function should then open the *same* file (`report.txt`) in read mode ('r') and read its entire contents.
5. Finally, the function should return the contents of the file as a single string.